WON IK JANG

green101630@gmail.com | GitHub | Research Portfolio

Education

Kyung Hee University *South Korea*
Biomedical Engineering & Electronic Engineering (Double Major) *Mar 2021 – Feb 2027 (expected)*
GPA: 4.26/4.50 **Entrance Scholarship:** Han Seung-Moo Scholarship (awarded to the top-ranked entrant)

Research Interests

Robot Learning • Imitation Learning • Robot Control • Sim-to-Real Transfer • Bimanual Manipulation

Research Experience

Kyung Hee University *South Korea*
Undergraduate Researcher | Advisor: Prof. Tae-Seong Kim *Sep 2025 – Present*

- Built and benchmarked a low-cost, dual-arm robot learning pipeline connecting ROS 2 teleoperation, Isaac Sim/Isaac Lab, demonstration collection, behavior cloning, and rollout evaluation.
- Developed Temporal-Context ACT (TC-ACT), an extension of ACT incorporating sampled observation histories; evaluated ACT and TC-ACT on bimanual pick-and-place and handover tasks.
- In a matched-reset simulation evaluation (1,250 trials per model), TC-ACT reached 82.72% handover success versus 74.32% for ACT; pick-and-place results were 93.12% and 93.44%, respectively.
- Deploying a SmolVLA policy trained on real-robot demonstrations for physical pick-and-place, with ongoing work focused on optimizing the robot control-loop frequency.
- Integrated Diffusion Policy with ROS 2 hardware using established PyTorch compilation and full-DDIM CUDA Graphs; verified ordered commands with live feedback and separately benchmarked model-only p99 latency (62.18 to 14.58 ms; RTX 3070, C128/DDIM10, 100 runs per mode).

Kyung Hee University Hospital at Gangdong *Seoul, South Korea*
Undergraduate Researcher, Department of Radiology | Advisor: Prof. Geon Ho Jahng *Dec 2023 – May 2025*

- Preprocessed MRI and clinical data for neuroimaging research on the relationship between renal function and cerebral blood flow; supported quantitative analysis and figure preparation.
- Contributed to a co-authored study subsequently published in *Chinese Medical Journal*.

Seoul National University – BiNEL *Seoul, South Korea*
Research Intern | Advisor: Prof. Sunghoon Kwon *Jun 2025 – Aug 2025*

- Participated in B-cell receptor (BCR) sequence recovery workflows and molecular biotechnology experiments, including PCR/qPCR condition assessment and optimization.
- Reviewed relevant research literature and assisted with experimental workflows for immune-repertoire analysis.

Publications

Rhee, H. Y., Choi, Y., Cha, J. M., Moon, J.-Y., Yoon, Y., **Jang, W. I.**, Park, S., Ryu, C.-W., Yang, D. M., & Jahng, G. H. (2026). "Brain–kidney connection: Impact on cerebral blood flow with estimated glomerular filtration rates in healthy brains." *Chinese Medical Journal*, 139(19), 3000–3002. doi:10.1097/CM9.0000000000004249.

Selected Engineering Projects

Cargo Stabilization Robot *Kyung Hee University*
Team Leader | Creative Innovation Competition *May 2026 – Nov 2026*

- Led a four-member team developing a mecanum-wheel mobile robot with an actively stabilized cargo tray.
- Integrated an STM32 microcontroller, IMU-based tilt estimation, and PD feedback control to compensate for platform roll and pitch.

Strength and EMG Measurement Device *Kyung Hee University*
Circuit Design and Hardware Prototyping *Apr 2024 – Nov 2024*

- Designed and assembled analog measurement circuitry for simultaneous muscle-strength and electromyography (EMG) acquisition.
- Supported hardware integration for real-time biomedical signal measurement and visualization.

Honors & Awards

Top Excellence Award, 2024 Creative Capstone Design Competition Oct 31, 2024
Silver Award, Sungkyunkwan University Consortium Competition Nov 8, 2024
President's Award, Korean Society for Engineering Education Nov 22, 2024

Technical Skills

Programming & ML: Python, C (basic), PyTorch
Robotics & Tools: ROS 2, Isaac Sim, Isaac Lab, MuJoCo, Docker